

Supplementary Appendix for Randomized Kolmogorov–Smirnov Calibration for Fitted Discrete Single-Cell Residual Diagnostics

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Statistics in Biosciences

A. One-Sample Crossing Formula

This appendix expands the volume calculation used in the proof of the Smirnov-Birnbaum-Tingey one-sided crossing formula. Work on the uniform scale. Let $U_{(1)} < \dots < U_{(n)}$ be the order statistics of independent $\text{Unif}(0, 1)$ variables and set $U_{(0)} = 0$, $U_{(n+1)} = 1$. On $[U_{(j)}, U_{(j+1)})$, the empirical distribution function equals j/n . Thus the first lower crossing at $t_j = \epsilon + j/n$ occurs when

$$U_{(i)} \leq \epsilon + \frac{i-1}{n}, \quad i = 1, \dots, j, \quad U_{(j+1)} > \epsilon + \frac{j}{n}.$$

For $j = 0$, this gives

$$\mathbb{P}\{\tau_n(\lambda) = \epsilon\} = (1 - \epsilon)^n.$$

For $j \geq 1$, the desired probability is $n!V_j$, where

$$V_j = \int_0^\epsilon \int_{u_1}^{\epsilon + \frac{1}{n}} \dots \int_{u_{j-1}}^{\epsilon + \frac{j-1}{n}} \int_{\epsilon + \frac{j}{n}}^1 \int_{u_{j+1}}^1 \dots \int_{u_{n-1}}^1 du_n \dots du_{j+2} du_{j+1} du_j \dots du_2 du_1.$$

Since $u_j \leq \epsilon + (j-1)/n < \epsilon + j/n \leq u_{j+1}$, this simplex separates into the product

$$V_j = \left(\int_0^\epsilon \int_{u_1}^{\epsilon + \frac{1}{n}} \dots \int_{u_{j-1}}^{\epsilon + \frac{j-1}{n}} du_j \dots du_2 du_1 \right) \times \left(\int_{\epsilon + \frac{j}{n}}^1 \int_{u_{j+1}}^1 \dots \int_{u_{n-1}}^1 du_n \dots du_{j+2} du_{j+1} \right).$$

The second factor is the ordinary ordered-simplex volume

$$\frac{(1 - \epsilon - j/n)^{n-j}}{(n-j)!}.$$

The first factor is the classical Birnbaum-Tingey simplex identity

$$\int_0^\epsilon \int_{u_1}^{\epsilon + \frac{1}{n}} \cdots \int_{u_{j-1}}^{\epsilon + \frac{j-1}{n}} du_j \cdots du_2 du_1 = \epsilon \frac{(\epsilon + j/n)^{j-1}}{j!}.$$

Therefore

$$\mathbb{P} \left\{ \tau_n(\lambda) = \epsilon + \frac{j}{n} \right\} = n! V_j = \epsilon \binom{n}{j} \left(\epsilon + \frac{j}{n} \right)^{j-1} \left(1 - \epsilon - \frac{j}{n} \right)^{n-j}.$$

Summing the mutually exclusive first-passage events gives

$$\mathbb{P}(D_n^- > \epsilon) = (1 - \epsilon)^n + \epsilon \sum_{j=1}^{\lfloor n(1-\epsilon) \rfloor} \binom{n}{j} \left(\epsilon + \frac{j}{n} \right)^{j-1} \left(1 - \epsilon - \frac{j}{n} \right)^{n-j}.$$

B. Exact two-sample recursion

Under $F = G$ and continuity, every pooled ordering of the n observations from the first sample and the m observations from the second sample is equally likely. At lattice point (i, j) the one-sided empirical difference is $i/n - j/m$. Counting paths that avoid the half-plane $i/n - j/m \geq \delta$ gives the recursion in Proposition 5. The total number of pooled orderings is $\binom{n+m}{n}$, so dividing the count of admissible paths by this total gives the finite-sample probability exactly.

C. Fixed-null crossing formulas

The following sections collect the finite-sample KS crossing derivations and fixed-null limiting identities used as background for the fitted residual diagnostics in the main text.

1 One Sample: The First Crossing Time

Let $X_1, \dots, X_n$ be a random sample with continuous distribution function F and let F_n be its empirical counterpart:

$$F_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i \leq t).$$

Define the normalized empirical process by

$$Z_n(t) = \sqrt{n}(F_n - F)(t)$$

and let D_n^+ and D_n^- denote the one-sided Kolmogorov-Smirnov statistics:

$$D_n^+ = \sup_t \frac{Z_n(t)}{\sqrt{n}} = \sup_t (F_n(t) - F(t)),$$

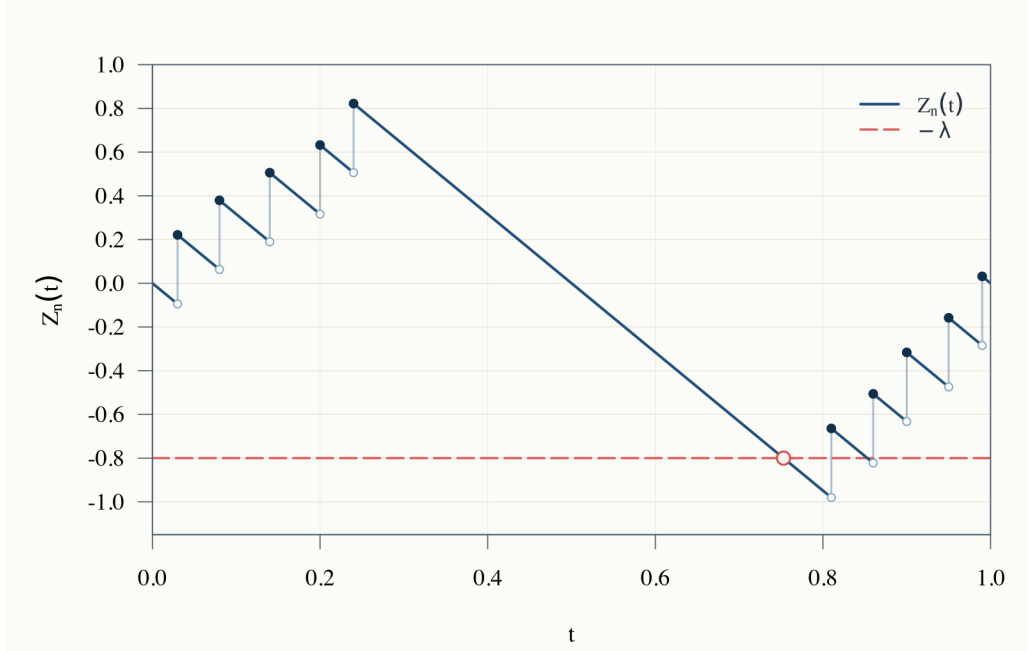


Figure 1: A one-sample empirical process on the uniform scale for $n = 10$. The blue path is $Z_n(t) = \sqrt{n}\{F_n(t) - t\}$; it drifts downward between observations and jumps upward at sample points. The red line is the lower boundary $-\lambda$ with $\lambda = 0.8$. The highlighted point is the first boundary crossing, reducing the KS tail calculation to the finite hitting-time question $t = \lambda/\sqrt{n} + j/n$.

$$D_n^- = \sup_t \frac{-Z_n(t)}{\sqrt{n}} = \sup_t (F(t) - F_n(t)).$$

Here the supremum becomes literal. Between observations the centered empirical process drifts downward; at an observation it jumps upward. A lower-boundary crossing therefore cannot occur everywhere on the line, but only at a small set of lattice points. The next theorem is the classical exact answer, written in this first-passage language.

Theorem 1 (Smirnov-Birnbaum-Tingey [1]). *Let $0 < \epsilon < 1$ and set $\lambda = \sqrt{n}\epsilon$. Define*

$$\tau_n(\lambda) = \inf_t \{t : Z_n(t) \leq -\lambda\}$$

as the first time that $Z_n(t)$ hits $-\lambda$. Then

$$\mathbb{P}(D_n^- > \epsilon) = (1 - \epsilon)^n + \epsilon \sum_{j=1}^{\lfloor n(1-\epsilon) \rfloor} \binom{n}{j} \left(\epsilon + \frac{j}{n}\right)^{j-1} \left(1 - \epsilon - \frac{j}{n}\right)^{n-j}. \quad (1)$$

On the uniform scale, the corresponding first-passage probabilities are

$$\mathbb{P}\{\tau_n(\lambda) = \epsilon\} = (1 - \epsilon)^n, \quad (2)$$

$$\mathbb{P}\left\{\tau_n(\lambda) = \epsilon + \frac{j}{n}\right\} = \epsilon \binom{n}{j} \left(\epsilon + \frac{j}{n}\right)^{j-1} \left(1 - \epsilon - \frac{j}{n}\right)^{n-j} \quad (3)$$

for $j = 1, \dots, \lfloor n(1 - \epsilon) \rfloor$.

Remark 2. The reduction to $F(t) = t$ is distribution-free: after the probability integral transform, the distributions of D_n^+ and D_n^- do not depend on F . On the original scale, the hitting locations are obtained by applying the generalized inverse F^{-1} to the uniform-scale values $\epsilon + j/n$.

Remark 3. The hitting time $\tau_n(\lambda)$ is discrete, although $\tau_n(\lambda) = +\infty$ may occur with positive probability. The process $Z_n(t)$ decreases between jump points; at each observation X_i , it jumps by $1/\sqrt{n}$. If $Z_n(t)$ does not cross below $-\lambda$ at $X_{(i)}$, the i -th order statistic, it must wait another $1/n$ in transformed time before crossing becomes possible. The first possible hitting time is $\epsilon = \lambda/\sqrt{n}$, corresponding to no observations below ϵ . Thus the equation

$$F_n(t) - t = -\epsilon$$

has only finitely many possible solutions, all of the form $\epsilon + j/n$.

Proof. It is enough to work on the uniform scale. Let $U_{(1)} < \dots < U_{(n)}$ be the order statistics of independent $\text{Unif}(0, 1)$ variables, with $U_{(0)} = 0$ and $U_{(n+1)} = 1$. On $[U_{(j)}, U_{(j+1)})$, the empirical distribution function is j/n . Thus the first crossing occurs at $t_j = \epsilon + j/n$ exactly when

$$U_{(i)} \leq \epsilon + \frac{i-1}{n} \quad (i = 1, \dots, j), \quad U_{(j+1)} > \epsilon + \frac{j}{n}.$$

The case $j = 0$ gives $(1 - \epsilon)^n$. For $j \geq 1$, the probability is $n!$ times the ordered-simplex volume cut out by these inequalities. The Birnbaum-Tingey simplex identity [1] gives this volume as

$$\frac{\epsilon}{j!(n-j)!} \left(\epsilon + \frac{j}{n} \right)^{j-1} \left(1 - \epsilon - \frac{j}{n} \right)^{n-j}.$$

Multiplying by $n!$ gives the displayed mass at $\epsilon + j/n$, and summing over $j = 0, \dots, \lfloor n(1 - \epsilon) \rfloor$ gives the tail probability. The ordered-simplex calculation is isolated in Appendix A above. $\square$

Remark 4 (Log-scale exact calculator). The preceding formula is a practical $O(n)$ evaluator, not only a closed form. Let $J = \lfloor n(1 - \epsilon) \rfloor$ and define

$$\ell_0 = n \log(1 - \epsilon),$$

and, for $j = 1, \dots, J$,

$$\ell_j = \log \epsilon + \log \binom{n}{j} + (j-1) \log \left(\epsilon + \frac{j}{n} \right) + (n-j) \log \left(1 - \epsilon - \frac{j}{n} \right),$$

with $\ell_j = -\infty$ whenever the last logarithm has zero argument. Then

$$\log \mathbb{P}(D_n^- > \epsilon) = \log \sum_{j=0}^J e^{\ell_j}.$$

This log-sum-exp form avoids overflow and cancellation in direct evaluation; an implementation is included in the source package.

2 Two Samples: Recursion Before Reflection

The two-sample statistic has the same crossing flavor, but the path now comes from the pooled ordering of two samples. With unequal sample sizes the boundary is weighted, so a direct reflection formula is no longer the cleanest object. The exact finite-sample calculation is instead a lattice recursion; reflection appears when the lattice is balanced. The next two propositions are the core of this calculation: they count the paths that avoid one wall, or both walls, for arbitrary n and m .

Let $X_1, \dots, X_n$ be a random sample with continuous distribution function F and let $Y_1, \dots, Y_m$ be another random sample with continuous distribution function G . Let F_n and G_m be the empirical counterparts of F and G :

$$F_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i \leq t), \quad G_m(t) = \frac{1}{m} \sum_{j=1}^m \mathbb{I}(Y_j \leq t).$$

Proposition 5 (Exact two-sample crossing recursion [4]). *Let*

$$D_{n,m}^+ = \sup_t \{F_n(t) - G_m(t)\}.$$

Under $F = G$, fix $\delta > 0$ and define $A_{0,0} = 1$. For $0 \leq i \leq n$ and $0 \leq j \leq m$, set $A_{i,j} = 0$ outside the rectangle and, for $(i, j) \neq (0, 0)$,

$$A_{i,j} = \mathbf{1} \left\{ \frac{i}{n} - \frac{j}{m} < \delta \right\} (A_{i-1,j} + A_{i,j-1}).$$

Then

$$\mathbb{P}(D_{n,m}^+ < \delta) = \frac{A_{n,m}}{\binom{n+m}{n}}, \quad \mathbb{P}(D_{n,m}^+ \geq \delta) = 1 - \frac{A_{n,m}}{\binom{n+m}{n}}.$$

Proof. Under the continuous null $F = G$, every pooled ordering of the n observations from the first sample and the m observations from the second sample is equally likely. A partial ordering with i observations from the first sample and j from the second sample has empirical difference $i/n - j/m$. Therefore the event $D_{n,m}^+ < \delta$ is exactly the event that every visited lattice point (i, j) satisfies $i/n - j/m < \delta$. The recursion counts admissible paths to (i, j) by adding the admissible paths entering from $(i-1, j)$ and $(i, j-1)$, and setting the count to zero whenever the boundary has already been hit. Since there are $\binom{n+m}{n}$ pooled orderings, division by this total gives the probability. $\square$

The same recursion can enforce both boundaries, giving an exact finite-sample calibrator for the ordinary two-sided statistic when n and m are unequal.

Proposition 6 (Two-wall recursion for unequal two-sample KS [4]). *Let*

$$D_{n,m} = \sup_t |F_n(t) - G_m(t)|.$$

Under $F = G$, fix $\delta > 0$ and define $B_{0,0} = 1$. For $0 \leq i \leq n$ and $0 \leq j \leq m$, set $B_{i,j} = 0$ outside the rectangle and, for $(i, j) \neq (0, 0)$,

$$B_{i,j} = \mathbf{1} \left\{ \left| \frac{i}{n} - \frac{j}{m} \right| < \delta \right\} (B_{i-1,j} + B_{i,j-1}).$$

Then

$$\mathbb{P}(D_{n,m} < \delta) = \frac{B_{n,m}}{\binom{n+m}{n}}, \quad \mathbb{P}(D_{n,m} \geq \delta) = 1 - \frac{B_{n,m}}{\binom{n+m}{n}}.$$

Proof. The preceding proof applies with the admissible half-plane replaced by the strip $|i/n - j/m| < \delta$. The recursion counts paths that remain in this strip from $(0, 0)$ to (n, m) ; division by the total number $\binom{n+m}{n}$ of pooled orderings gives the probability. $\square$

Only grid values of $|i/n - j/m|$ need be checked, so the same recursion also returns exact critical values for any desired level.

When $n = m$, write G_n for the second empirical distribution. The weighted boundary becomes a constant level for a simple walk, and the recursion collapses to the familiar reflection count.

The balanced case gives the classical Gnedenko-Korolyuk reflection formula.

Theorem 7 (Balanced reflection formula [4]). *Let $D_{n,n}^+$ be the one-sided Kolmogorov-Smirnov statistic:*

$$D_{n,n}^+ = \sup_t (F_n(t) - G_n(t)).$$

If $F = G$, then for $r = 1, \dots, n$,

$$\mathbb{P}\left(D_{n,n}^+ \geq \frac{r}{n}\right) = \frac{\binom{2n}{n-r}}{\binom{2n}{n}}. \quad (4)$$

Proof. Encode each pooled ordering of the X and Y samples as a path with $2n$ steps: an X -observation contributes $+1$, and a Y -observation contributes -1 . After $2n$ steps, the path returns to 0. The total number of such paths is

$$\binom{2n}{n}.$$

The event $\{D_{n,n}^+ \geq r/n\}$ is equivalent to the event that the path hits level r . By the reflection principle [4], reflecting the path after its first hit of r maps such paths bijectively to paths that end at $2r$. Thus the number of crossing paths is

$$\binom{2n}{n+r} = \binom{2n}{n-r}.$$

Since all pooled orderings are equally likely under the null,

$$\mathbb{P}\left(D_{n,n}^+ \geq \frac{r}{n}\right) = \frac{\binom{2n}{n-r}}{\binom{2n}{n}}.$$

$\square$

Corollary 8 (Feller limit [5]). *For fixed $r > 0$, if $n \rightarrow \infty$, then*

$$\mathbb{P}\left(D_{n,n}^+ \geq \frac{r}{\sqrt{n}}\right) \rightarrow e^{-r^2}. \quad (5)$$

Proof. Let $k_n = \lceil r\sqrt{n} \rceil$. Since $D_{n,n}^+$ takes values on the grid $\{0, 1/n, \dots, 1\}$, the preceding theorem gives

$$\mathbb{P}\left(D_{n,n}^+ \geq \frac{r}{\sqrt{n}}\right) = \frac{n!n!}{(n-k_n)!(n+k_n)!}.$$

Stirling's formula, with $k_n/n \rightarrow 0$ and $k_n^2/n \rightarrow r^2$, gives the limit e^{-r^2} . $\square$

D. Uniform bounds and limiting laws

The two-sided one-sample KS statistic is defined as

$$D_n = \sup_t |F_n(t) - F(t)|.$$

The one-sided story keeps track of the first lower crossing. The two-sided statistic asks the path to hit either wall. Exact two-sided formulas exist, but for many applications the right object is the sharp envelope: a bound that forgets the particular hitting location while preserving the correct exponential scale.

Theorem 9 (Dvoretzky-Kiefer-Wolfowitz-Massart [3, 7]). *For every distribution function F and every $\epsilon > 0$,*

$$\mathbb{P}(D_n > \epsilon) \leq 2e^{-2n\epsilon^2}. \quad (6)$$

Massart [7] proved that the constant 2 is sharp; see also Kosorok [6] for a modern empirical-process treatment, including distribution functions with jumps. The connection with the crossing formula above is that the one-sided event is exactly the finite sum

$$\mathbb{P}(D_n^- > \epsilon) = (1 - \epsilon)^n + \epsilon \sum_{j=1}^{\lfloor n(1-\epsilon) \rfloor} \binom{n}{j} \left(\epsilon + \frac{j}{n}\right)^{j-1} \left(1 - \epsilon - \frac{j}{n}\right)^{n-j}.$$

Massart's argument controls this crossing sum, and its upper analogue, at the exponential scale

$$\exp(-2n\epsilon^2).$$

Thus the DKWM inequality is not a separate phenomenon in this narrative: it is the sharp uniform bound obtained after the exact first-passage calculation is compressed into an exponential tail.

Remark 10 (Numerical scale). The exact calculation is small enough to be used directly. For example, when $n = 100$, the one-sided crossing formula gives $\mathbb{P}(D_n^- > 0.10) = 0.1266$ and $\mathbb{P}(D_n^- > 0.12) = 0.0517$, whereas the two-sided DKWM envelopes are 0.2707 and 0.1123. For the unequal two-sample recursion, with $(n, m) = (100, 80)$ and $\delta = 0.12$, the exact values are $\mathbb{P}(D_{n,m}^+ \geq \delta) = 0.2570$ and $\mathbb{P}(D_{n,m} \geq \delta) = 0.5056$. These values are reproduced by the exact-crossing script included in the source package.

Two-sample limit

For the two-sided two-sample case, define

$$D_{n,m} = \sup_t |F_n(t) - G_m(t)|.$$

Under the null hypothesis $F = G$, its limiting distribution is

$$\lim_{n,m \rightarrow \infty} \mathbb{P} \left(\sqrt{\frac{nm}{n+m}} D_{n,m} > \lambda \right) = 2 \sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2 \lambda^2}. \quad (7)$$

The two-wall recursion in Section 2 is exact but finite. The display above is what remains when the lattice is scaled and allowed to converge: the path becomes a Brownian bridge and the alternating exponential series replaces path counting. This limit can be shown via Donsker's theorem or empirical process theory [6]. For richer suprema over function classes, the limiting distribution may also depend on the common distribution $F = G$.

E. Geometric and crossing-complex interpretation

The following material gives the optional geometric and topological interpretation of the fitted empirical-process and finite path-counting views.

3 Geometry and Topology of Crossing Calibration

The fitted-null and finite-sample calculations have a common structure. In regular score-based models, parameter fitting removes directions from a stochastic process. At finite sample size, boundary avoidance restricts paths to an admissible region of a lattice. These are geometric and topological statements, but they remain tied to the calibration problem.

3.1 Model-manifold projection

Let $\mathcal{M} = \{F_\theta : \theta \in \Theta\}$ be a smooth parametric model. At F_{θ_0} , the tangent directions are the derivatives $\dot{F}_{\theta_0}(t) = \partial F_\theta(t) / \partial \theta|_{\theta=\theta_0}$. For maximum likelihood estimation, the fitted empirical process can be written as a Brownian bridge with its score-tangent component removed.

Proposition 11 (Tangent projection form). *Assume a regular parametric model with score s_{θ_0} and Fisher information I_{θ_0} . Let B_F denote the F_{θ_0} -Brownian bridge and let*

$$S = \int s_{\theta_0}(x) dB_F(x).$$

Then the fitted-null empirical process has the limit

$$Z(t) = B_F(t) - \dot{F}_{\theta_0}(t)^T I_{\theta_0}^{-1} S.$$

Thus fitting replaces the simple-null Brownian bridge by a projected Gaussian process whose covariance is the Brownian-bridge covariance minus the component explained by the score tangent space.

Proof. For maximum likelihood estimation,

$$\sqrt{n}(\hat{\theta} - \theta_0) = I_{\theta_0}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n s_{\theta_0}(X_i) + o_{\mathbb{P}}(1).$$

Substituting this expansion into the first-order Taylor expansion of $F_{\hat{\theta}}(t)$ gives

$$\sqrt{n}\{F_n(t) - F_{\hat{\theta}}(t)\} = \sqrt{n}\{F_n(t) - F_{\theta_0}(t)\} - \dot{F}_{\theta_0}(t)^T \sqrt{n}(\hat{\theta} - \theta_0) + o_{\mathbb{P}}(1).$$

The joint empirical-process limit of the first term and the score sum yields the display. $\square$

This proposition is the differential-geometric version of the calibration warning: a fitted simulator defines a model manifold, and the diagnostic path is the empirical fluctuation after projection along the tangent directions used by the fit. The derivative $\dot{G}_{\theta_0}$ in the main text's plug-in randomized bridge result is the corresponding tangent map after randomized residualization.

3.2 Crossing complexes for exact recursions

The two-sample recursion also has a topological reading. Let $\mathcal{G}_{n,m}$ be the directed grid graph on vertices (i, j) , $0 \leq i \leq n$, $0 \leq j \leq m$, with directed edges $(i, j) \rightarrow (i+1, j)$ and $(i, j) \rightarrow (i, j+1)$. For $\delta > 0$, define the two-wall admissible vertex set

$$V_{\delta} = \left\{ (i, j) : \left| \frac{i}{n} - \frac{j}{m} \right| < \delta \right\}.$$

The induced directed subgraph $\mathcal{K}_{\delta}$ is the crossing complex. As δ varies, these subgraphs form a filtration of admissible regions.

Proposition 12 (Directed path-count filtration). *Proposition 6 counts the number of directed paths from $(0, 0)$ to (n, m) in the crossing complex $\mathcal{K}_{\delta}$. Equivalently, if $B_{i,j}(\delta)$ denotes the number of admissible directed paths from $(0, 0)$ to (i, j) , then*

$$B_{i,j}(\delta) = \mathbf{1}\{(i, j) \in V_{\delta}\} \{B_{i-1,j}(\delta) + B_{i,j-1}(\delta)\},$$

with $B_{0,0}(\delta) = 1$. Therefore

$$\mathbb{P}(D_{n,m} < \delta) = \frac{B_{n,m}(\delta)}{\binom{n+m}{n}}.$$

Proof. Every pooled ordering corresponds to exactly one directed monotone path in $\mathcal{G}_{n,m}$, and the empirical difference at a partial ordering is $i/n - j/m$. The event $D_{n,m} < \delta$ is exactly the event that the associated path remains inside V_{δ} . The displayed recursion is the directed incidence recursion for counting paths in the induced subgraph. $\square$

The zero-one question of whether $(0, 0)$ and (n, m) are connected inside $\mathcal{K}_{\delta}$ is a directed analogue of zeroth persistent connectivity. The KS distribution needs a sharper invariant: not only whether a terminal path exists, but how many admissible paths exist under the uniform measure on pooled orderings. Thus the exact KS recursion is a path-count refinement of a topological crossing filtration.

F. Deferred empirical-process proofs

This section gives the derivations deferred from the main text. All notation follows the corresponding theorem and corollaries in the main manuscript.

F.1 Classical composite-null expansion

Write P_n for the empirical measure and P_θ for the law under F_θ . Asymptotic linearity gives

$$\sqrt{n}(\hat{\theta} - \theta) = n^{-1/2} \sum_{i=1}^n \psi_\theta(X_i) + o_{\mathbb{P}_\theta}(1).$$

Uniform differentiability of $F_\theta(t)$ in θ gives, uniformly in t ,

$$\sqrt{n}\{F_{\hat{\theta}}(t) - F_\theta(t)\} = \frac{\partial F_\theta(t)^T}{\partial \theta} \sqrt{n}(\hat{\theta} - \theta) + o_{\mathbb{P}_\theta}(1).$$

Therefore

$$Z_n(t, \hat{\theta}) = \sqrt{n}(P_n - P_\theta)\mathbf{1}\{X \leq t\} - \frac{\partial F_\theta(t)^T}{\partial \theta} n^{-1/2} \sum_{i=1}^n \psi_\theta(X_i) + o_{\mathbb{P}_\theta}(1)$$

uniformly over the KS index t . The assumed joint convergence in $\ell^\infty(\mathcal{T}) \times \mathbb{R}^k$ and the displayed uniform remainder imply weak convergence of this linear combination to a mean-zero Gaussian limit. Computing its covariance gives the covariance display in the main text. The continuous mapping theorem applied to the supremum norm gives convergence of the tail probabilities at continuity points of the limiting law.

F.2 Plug-in randomized bridge

Set $T_\theta(x, v) = T_{F_\theta}(x, v)$. At the true parameter, the known-margin randomized-transform proposition gives $T_{\theta_0}(X, V) \sim \text{Unif}(0, 1)$, so $G_{\theta_0}(u) = u$. The estimator expansion in the corollary is the asymptotic-linearity condition in the residual empirical-process theorem, with influence function $\psi_{\theta_0}(X)$ that does not depend on the auxiliary randomizer V . The remaining assumptions are precisely the Donsker, plug-in-remainder, local-centering-linearity, and joint-convergence conditions of that theorem, with $H_\theta = G_\theta$. Substitution yields

$$\alpha_n(u) \Rightarrow B(u) + \dot{G}_{\theta_0}(u)^T Z_\theta$$

in $\ell^\infty(0, 1)$.

F.3 Refitted simulator bootstrap

Apply the residual empirical-process theorem conditionally on the observed data. Let $P_{\hat{\theta}}$ be the conditional law of a bootstrap pair (X^*, V^*) , with $X^* \sim F_{\hat{\theta}}$ and $V^* \sim \text{Unif}(0, 1)$, and

let P_n^* be the empirical measure of the pairs (X_i^*, V_i^*) . The bootstrap assumptions imply conditional asymptotic linearity,

$$\sqrt{n}(\hat{\theta}^* - \hat{\theta}) = n^{-1/2} \sum_{i=1}^n \psi_{\hat{\theta}}(X_i^*) + o_{\mathbb{P}^*}(1),$$

where $o_{\mathbb{P}^*}(1)$ denotes convergence in conditional bootstrap probability. They also give the bootstrap analogue of the plug-in empirical-process expansion. Writing $\dot{G}_{\hat{\theta}}^*(u)$ for the derivative at $\eta = \hat{\theta}$ of $\eta \mapsto P_{\hat{\theta}}\{T_{F_{\eta}}(X^*, V^*) \leq u\}$, this expansion is

$$\begin{aligned} \alpha_n^*(u) &= \sqrt{n} \{P_n^* \mathbf{1}(T_{F_{\hat{\theta}^*}}(X^*, V^*) \leq u) - u\} \\ &= \sqrt{n}(P_n^* - P_{\hat{\theta}}) \mathbf{1}\{T_{F_{\hat{\theta}}}(X^*, V^*) \leq u\} + \dot{G}_{\hat{\theta}}^*(u)^T \sqrt{n}(\hat{\theta}^* - \hat{\theta}) + o_{\mathbb{P}^*}(1), \end{aligned}$$

uniformly in u , in conditional probability. Thus the bootstrap process has the same first-order Brownian-bridge term and the same first-order estimation term as the original fitted residual process, with $\hat{\theta}$ playing the role of the true parameter. Conditional weak convergence implies convergence of the conditional law of the supremum at continuity points of the limiting law. If $\hat{\theta}$ is held fixed in the bootstrap sample, the expansion lacks the estimation term $\dot{G}_{\theta_0}(u)^T Z_{\theta}$ and generally calibrates a different process.

G. Public IMvigor210 baseline-PBMC analysis

The clinical example uses public baseline-PBMC integer UMI counts from 10 IMvigor210 patients (GEO GSE145281): five responders and five nonresponders [10]. Response labels are descriptive; neither treatment efficacy nor response association is tested. The phase 2 cohort is registered as NCT02108652 [9].

The matrices contain 26,749 candidate cells and 30,727 RefSeq features. Our independent QC retains 24,134 cells with at least 500 UMIs, 200–6000 detected genes, and mitochondrial fraction at most 0.20. The source-motivated 10-gene panel was fixed before residual fitting. Each working margin uses library-size offsets, 10 exposure-adjusted patient rates, and a shared NB2 dispersion. For fitted means $\hat{\mu}_i$, the dispersion estimate is

$$\hat{\alpha} = \max \left\{ 10^{-3}, \frac{\sum_i \{(Y_i - \hat{\mu}_i)^2 - Y_i\}}{\sum_i \hat{\mu}_i^2} \right\}, \quad \hat{r} = \min\{1000, \max(0.05, 1/\hat{\alpha})\}.$$

The statistic is $\max_j \sqrt{n_j} D_j$, where D_j is the patient-wise randomized-PIT KS distance. Both references use the same $B = 9,999$ simulated count matrices. The fixed reference retains the observed parameters; the refitted reference re-estimates all rates and the dispersion. Both generate fresh residual randomizers and use plus-one Monte Carlo p -values.

The exploratory extension replaces shared r_g by patient-specific r_{jg} , using the same moment equation within each patient and the same truncation limits. The means, patient indicators, QC decisions, offsets, and full 10-gene panel are unchanged. This increases the number of fitted parameters per gene from 11 to 20. The extension was specified in response to the shared-model discrepancies and evaluated once, without searching over models until the diagnostic passed. Each candidate is simulated from its own fitted distribution and

fully refitted in all $B = 9,999$ replicates. Holm adjustment is over the 10 genes within each candidate model, not over model selection. The comparison is exploratory rather than a confirmatory test of the superiority of the extension. Fitted log scores below are in-sample summaries, not estimates of out-of-sample predictive performance.

For zero-based gene index g , the observed NumPy generator uses seed $20260822 + 1009g$. Both models use the same observed randomizers. The bootstrap generator uses `SeedSequence([seed, 20260907])`, keeping observed and simulated randomization streams separate. A failure stops the analysis rather than removing a replicate. Four gene-model workers are used; the per-task CSV timings cover the bootstrap loop and stored-summary calculation, not data loading.

The reference assumes conditional cell independence and fixes QC decisions and library-size offsets. Rejection can reflect these sampling assumptions as well as the count margin. The minimum reportable value is $1/10000 = 0.0001$. Appendix H.4 defines the conditional Monte Carlo intervals and their limitations.

Table 1: Shared-dispersion IMvigor210 baseline-PBMC diagnostic ($B = 9,999$). Detection is the retained-cell nonzero fraction, r is the fitted NB size, and S is the maximum patient-stratified KS statistic. “Fixed” and “refit” distinguish the two bootstrap reference procedures.

Gene	Detection	r	S	Fixed 95%	Refit 95%	Fixed p	Refit p	Holm p
CXCL8	0.348	0.362	8.768	1.723	1.905	0.0001	0.0001	0.0010
B2M	0.993	6.850	8.849	1.715	1.340	0.0001	0.0001	0.0010
CD74	0.665	0.318	12.855	1.721	2.260	0.0001	0.0001	0.0010
HLA-A	0.879	3.727	3.515	1.721	1.340	0.0001	0.0001	0.0010
HLA-C	0.923	4.365	3.017	1.711	1.335	0.0001	0.0001	0.0010
HLA-DQA1	0.026	0.141	1.140	1.721	1.709	0.7996	0.7785	0.7785
HLA-DQB1	0.247	0.232	8.248	1.723	1.791	0.0001	0.0001	0.0010
HLA-DRB1	0.421	0.344	8.779	1.729	1.889	0.0001	0.0001	0.0010
TAP1	0.092	1.229	1.520	1.723	1.646	0.1824	0.1188	0.3564
TAP2	0.037	1.263	1.526	1.717	1.705	0.1636	0.1470	0.3564

All 199,980 refits succeeded. The same seven genes are rejected after within-model Holm correction under both models; the extension’s adjusted values range from 0.0010 to 0.0308. Changes in S and in fitted log score are not uniformly favorable. In particular, S increases for CD74 and HLA-C, whereas the log score decreases for HLA-DQB1. The moment estimators do not maximize likelihood, so adding dispersion parameters need not increase fitted likelihood. Allowing patient-specific dispersion is insufficient to remove the seven discrepancies. Their causes remain unidentified, and these conditional diagnostics do not test the published clinical biomarker conclusions.

H. Simulation and implementation details

H.1 Classical composite-null comparison

The normal and exponential comparison uses $n = 100$, 2000 Monte Carlo samples, 399 bootstrap samples per fitted data set, and nominal level 5%. The classical cutoff is $1.358/\sqrt{n}$.

Table 2: Exploratory patient-specific-dispersion comparison. S_0 and S_1 are observed statistics under shared and patient-specific dispersions, respectively. The cutoff, refitted p -value, and within-model Holm value refer to the extension ($B = 9,999$). $\Delta\ell$ is the difference in mean fitted log score per cell (extension minus baseline); positive values indicate a higher in-sample score, not demonstrated predictive improvement.

Gene	S_0	S_1	Refit 95%	Refit p	Holm p	$\Delta\ell$
CXCL8	8.768	5.403	3.226	0.0077	0.0308	0.034098
B2M	8.849	3.275	1.567	0.0023	0.0138	0.033575
CD74	12.855	14.004	4.064	0.0001	0.0010	0.001244
HLA-A	3.515	3.010	1.287	0.0003	0.0027	0.006357
HLA-C	3.017	3.796	1.312	0.0003	0.0027	0.003125
HLA-DQA1	1.140	1.119	1.702	0.8033	0.8033	0.001226
HLA-DQB1	8.248	4.895	2.184	0.0010	0.0070	-0.006441
HLA-DRB1	8.779	5.122	3.169	0.0058	0.0290	0.002032
TAP1	1.520	1.437	1.638	0.1868	0.3873	0.000276
TAP2	1.526	1.547	1.699	0.1291	0.3873	0.000307

Table 3: Conditional Monte Carlo uncertainty for the IMvigor210 refitted tails ($B = 9,999$ for each model). K_0, K_1 denote exceedance counts under shared and patient-specific dispersions. The intervals are marginal 95% Clopper–Pearson intervals for the conditional tail probabilities.

Gene	K_0	Interval, shared	K_1	Interval, patient-specific
CXCL8	0	[0, 0.000369]	76	[0.005993, 0.009504]
B2M	0	[0, 0.000369]	22	[0.001379, 0.003329]
CD74	0	[0, 0.000369]	0	[0, 0.000369]
HLA-A	0	[0, 0.000369]	2	[0.000024, 0.000722]
HLA-C	0	[0, 0.000369]	2	[0.000024, 0.000722]
HLA-DQA1	7784	[0.770208, 0.786585]	8032	[0.795351, 0.811033]
HLA-DQB1	0	[0, 0.000369]	9	[0.000412, 0.001708]
HLA-DRB1	0	[0, 0.000369]	57	[0.004320, 0.007380]
TAP1	1187	[0.112434, 0.125213]	1867	[0.179123, 0.194498]
TAP2	1469	[0.140029, 0.154006]	1290	[0.122501, 0.135741]

The implementation is `scripts/ks_composite_simulation.py`, run with its default settings and seed 20260520. Each bootstrap sample is fitted anew before its KS statistic is computed; the critical value is the empirical 95th percentile, using the upper order statistic when interpolation would otherwise be required.

H.2 Discrete-margin calibration

The negative-binomial sample-size experiment uses mean $\mu = 6$ and size $r = 2.5$, with variance $\mu + \mu^2/r$. For each sample, the moment fit sets $\hat{\mu} = \max(\bar{X}, 10^{-6})$. With unbiased sample variance s^2 , the fitted size is 1000 when $s^2 \leq 1.02\hat{\mu}$; otherwise it is $\hat{\mu}^2/(s^2 - \hat{\mu})$ truncated to $[0.05, 1000]$. The same rule is used in each refit. The alternative replaces each generated

count by zero, independently, with probability 0.15; thus it is a mixture of the null count distribution and a point mass at zero, not a 15% increase in its existing zero probability.

For $n = 250, 1,000$, and $10,000$, the null Monte Carlo sample sizes are 400, 250, and 300; the power sample sizes are 200, 125, and 100; and the bootstrap sizes are 199, 149, and 99, respectively. Known-margin and naive tests use $1.358/\sqrt{n}$. The bootstrap tests compare the observed KS distance to the empirical 95th percentile with the upper-order-statistic convention described in H.1. The fixed and refitted references share simulated count samples but use fresh independent residual randomizers. Refitting also re-estimates both negative-binomial parameters.

The refitted null rejection rates, with 95% Wilson Monte Carlo intervals, are 0.062 [0.043, 0.091], 0.040 [0.022, 0.072], and 0.040 [0.023, 0.069], respectively. Wilson intervals retain nonzero width at zero or complete rejection. They quantify uncertainty from the finite number of simulated data sets, not uncertainty in a model parameter.

The sample-size, copula, and lightweight scalability experiments are implemented in `scripts/ks_scdesign_extended_simulations.py`. Its seed is 20260609; the numerical output is in `results/ks_scdesign_extended_results.txt`. The copula experiment uses $n = 240$ cells in each of two independent samples and 1000 Monte Carlo samples. Its gene-wise statistics are two-sample KS distances with known count margins; the dependence comparator uses the difference of Fisher-transformed residual Pearson correlations with standard error $\{2/(n - 3)\}^{1/2}$. This is an approximate correlation reference, not a refitted copula test. The integer rejection counts and Wilson intervals are also preserved in `results/synthetic_rejection_counts.csv`. The 10-gene scalability experiment processes one gene at a time; its memory summary is the analytic three-array baseline, not measured peak memory.

H.3 Package examples and computational environment

The scGTM demonstration reads `scGTM/Demo/simu_nb_scGTM_input.csv` from commit `f26946dbb1531e273bed54f6eb6d51c6a2a5a408` of the [scGTM repository](#). It uses 500 observations and the first 10 genes of the supplied count matrix and fixed pseudotime. The repository does not identify the generating mechanism of this demonstration matrix; it is not used as independent biological validation or as an experiment with a verified data-generating null. Poisson and negative-binomial trends are fitted with 60 iterations of the original particle-swarm optimizer, not the CSO-MA optimizer studied in [2]. Complete gene-level and aggregate summaries are in `results/scgtm_example/`; the source commit, seed, and command are recorded in `results/ks_scdesign_extended_results.txt`.

The `scDesign3` object `example_count` is the 100-gene, 2,087-cell mouse pancreatic subset described in the [scDesign3 introduction tutorial](#). The separate bundled `example_sce` contains 10 genes and 1,289 human preimplantation embryo cells from E-MTAB-3929 [8]. The verification script `scripts/verify_scdesign3_provenance.R` matches all 1,289 cell identifiers and developmental-stage labels to the public SDRF sample file. Its `cell_type` field encodes embryonic days 3–7. The supplied counts and pseudotimes are used as distributed; the original preprocessing and pseudotime estimation are not repeated.

Conditional marginal fits call `scDesign3::fit_marginal()`, the package function for fitting per-gene regression models, with `family_use="nb"`, `mu_formula="s(pseudotime, bs='cr', k=5)"`, `sigma_formula="1"`, and `simplify=TRUE`.

The pancreatic calibration uses the first 10 genes in the supplied ordering and $B = 1,999$ simulated count vectors per gene. For each vector, the same conditional mean and dispersion regressions are fitted anew, fresh PIT randomizers are drawn, and the KS distance is recomputed. The observed randomization seed is $20260815 + 1009(g - 1)$ for original gene position g ; the bootstrap stream starts at that seed plus 100000. Separating these streams keeps the observed statistic and the initial bootstrap replicates unchanged when B or the worker order changes. All 19,990 fits succeeded. The implementation stops if any fit fails rather than silently discarding failed replicates. Computation uses four gene-level workers and one fitting core per worker. Timings in the gene-level CSV include the observed fit and the bootstrap loop, not data loading.

H.4 Monte Carlo reporting and stored records

For observed statistic S and independent bootstrap statistics $S_1^*, \dots, S_B^*$, let $K = \sum_b \mathbf{1}\{S_b^* \geq S\}$. The reported value is $(K + 1)/(B + 1)$. Conditional on the fitted generator, observed statistic, and fixed design, K is binomial with tail probability $q = \Pr^*(S^* \geq S)$. We report exact two-sided 95% Clopper–Pearson intervals for q , using beta quantiles and endpoints 0 or 1 when appropriate. These are marginal Monte Carlo intervals, not simultaneous confidence intervals, biological uncertainty intervals, or an assessment of bootstrap approximation error. Zero exceedances do not establish a zero tail probability. Holm correction is applied to the plus-one values within each specified 10-gene panel; it does not account for exploratory model selection.

Table 4: Pancreatic bootstrap exceedances and conditional Monte Carlo uncertainty, with $B = 1,999$ for every gene. The interval estimates the conditional tail probability q , not the plus-one estimator itself.

Gene	K	Refit p	95% Monte Carlo interval for q
Pyy	4	0.0025	[0.000545, 0.005115]
Iapp	3	0.0020	[0.000310, 0.004380]
Chgb	244	0.1225	[0.108023, 0.137218]
Rbp4	0	0.0005	[0, 0.001844]
Spp1	142	0.0715	[0.060160, 0.083188]
Chga	3	0.0020	[0.000310, 0.004380]
Cck	576	0.2885	[0.268363, 0.308549]
Ins1	303	0.1520	[0.136125, 0.168056]
Nnat	0	0.0005	[0, 0.001844]
Ins2	122	0.0615	[0.050937, 0.072434]

The scDesign3 pancreatic refits and Appendix G clinical outputs are in `results/scdesign3_example/` and `results/genentech_invigor210/`, respectively. Per-gene RDS or NPZ records preserve every bootstrap statistic and the observed residuals. CSV files give exceedance counts, conditional Monte Carlo intervals, cutoffs, seeds, elapsed times, and failed-fit counts. The repository README lists the reproduction commands, and `results/reproducibility_session_info.txt` records package versions and the computational environment.

The public code and retained numerical evidence are distributed as [release v1.0.0](#) of the [companion repository](#). The release includes the clean manuscript, this supplement, and a per-file SHA-256 manifest; its README gives build and verification commands.

H.5 Refitted conditional-margin simulation

This experiment uses `scDesign3`'s actual conditional-margin fitting function, not least squares on log counts. Pseudotimes are fixed at $t_i = (i - 0.5)/n$, and independent counts have NB mean μ_i and size $r = 3$. The three $n = 240$ scenarios are: a log-linear null with $\log \mu_i = 1 + 1.2t_i$; a periodic alternative $\log \mu_i = 1 + 1.2t_i + 0.8 \sin(2\pi t_i)$ fitted by the log-linear model; and that periodic null fitted using the specified sine and cosine terms. The fourth scenario uses $n = 2,087$ and a fixed $k = 5$ cubic-regression-spline null. Its log mean is $1 + 1.2t$ plus the least-squares projection of $0.8 \sin(2\pi t)$ onto the natural-cubic basis returned by `mgcv::smoothCon`. This generating curve is fixed before the count samples are drawn and lies in the fitted spline space.

Every scenario uses 500 independent outer samples and $B = 199$ bootstrap replicates per outer sample, with nominal level 0.05. Each fit calls `scDesign3::fit_marginal` with `family_use="nb"`, `sigma_formula="1"`, `usebam=FALSE`, and `simplify=TRUE`. The selected implementation fits an NB regression using `mgcv::gam`; both the mean parameters and NB size are estimated. For the spline scenario, the formula is the same `s(pseudotime, bs='cr', k=5)` used for the pancreatic margins; smoothing is also re-estimated. A prepared data container avoids repeating object construction, and its fitted means and dispersions have been checked against the package's `construct_data/extract_para` route.

Let $Q_1, \dots, Q_4$ be the fixed pseudotime quartiles, with sizes n_a , and let D_{ab} be the two-sample KS distance between their residuals. The two scalar statistics are

$$S_{\text{marginal}} = \sqrt{n} \sup_u |F_{n,U}(u) - u|, \quad S_{\text{conditional}} = \max_{a < b} \sqrt{\frac{n_a n_b}{n_a + n_b}} D_{ab}.$$

The reference distribution calibrates the entire maximum, rather than applying six unadjusted tests. Each bootstrap sample preserves the fixed design and is drawn from the observed fit. A complete refit, fresh PIT randomizers, and both statistics are recomputed. The fixed-parameter comparator shares simulated counts and randomizers and omits only refitting. Rejection uses $(1 + K)/(B + 1) \leq 0.05$. The two statistics are reported as separate diagnostic targets, not combined into an additional rejection rule.

For scenario index $s = 1, \dots, 4$ and trial $m = 1, \dots, 500$, the observed seed is $20260908 + 100000s + m$; the bootstrap seed adds 10^7 . Observed counts and randomizers therefore do not change with B or worker order. Invalid or nonconvergent fits stop the run; no failed replicates are dropped. Four outer-sample workers are used, with one fitting core each. The results in `results/conditional_calibration/` retain every observed count vector, residual vector, fitted parameter summary, bootstrap statistic, seed, timing, and warning count. The implementation is `scripts/ks_conditional_calibration.R` and its numerical verification is `scripts/verify_conditional_calibration.R`. All 2,000 observed fits and 398,000 bootstrap refits converged, with no recorded warnings. The full four-scenario run took 3,122.6 seconds of wall time on the reported four-worker environment; this is not a full-simulator runtime benchmark.

Rejection-rate intervals are exact two-sided 95% Clopper–Pearson intervals over the 500 outer samples. At a true rate of 0.05, the Monte Carlo standard error is approximately 0.0097. This experiment assesses the specified fixed-design regressions at finite sample size; it does not establish consistency for selected gene panels, inferred trajectories, copula fits, or arbitrary scDesign3 configurations.

Table 5: Full conditional-margin simulation results. The generating and fitted mean specifications are separated by a slash. Each row uses 500 outer samples and 199 inner bootstrap replicates per sample. Intervals are exact 95% binomial intervals for the outer rejection probability.

Generating / fitted	Statistic	Reference	n	Rejections	Rate	95% interval
Linear / linear	Marginal	Fixed	240	1/500	0.002	[0.000;0.011]
Linear / linear	Marginal	Refitted	240	32/500	0.064	[0.044;0.089]
Linear / linear	Conditional	Fixed	240	11/500	0.022	[0.011;0.039]
Linear / linear	Conditional	Refitted	240	26/500	0.052	[0.034;0.075]
Periodic / linear	Marginal	Fixed	240	1/500	0.002	[0.000;0.011]
Periodic / linear	Marginal	Refitted	240	42/500	0.084	[0.061;0.112]
Periodic / linear	Conditional	Fixed	240	437/500	0.874	[0.842;0.902]
Periodic / linear	Conditional	Refitted	240	462/500	0.924	[0.897;0.946]
Periodic / periodic	Marginal	Fixed	240	0/500	0.000	[0.000;0.007]
Periodic / periodic	Marginal	Refitted	240	24/500	0.048	[0.031;0.071]
Periodic / periodic	Conditional	Fixed	240	3/500	0.006	[0.001;0.017]
Periodic / periodic	Conditional	Refitted	240	32/500	0.064	[0.044;0.089]
Spline / spline	Marginal	Fixed	2087	0/500	0.000	[0.000;0.007]
Spline / spline	Marginal	Refitted	2087	30/500	0.060	[0.041;0.085]
Spline / spline	Conditional	Fixed	2087	1/500	0.002	[0.000;0.011]
Spline / spline	Conditional	Refitted	2087	32/500	0.064	[0.044;0.089]

H.6 Hardware, computational workload, and timing conventions

The conditional experiment in H.5 was run on an Apple M1 Max MacBook Pro (MacBookPro18,2), with 10 CPU cores (eight performance and two efficiency cores), 64 GiB memory, and macOS 26.5.2 (build 25F84). The hardware inventory was read from the analysis host on 9 September 2026, following the conditional run. The R environment is R 4.5.3, scDesign3 1.8.0, mgcv 1.9-4, SingleCellExperiment 1.32.0, and gamlss.dist 6.1-1, with the R framework BLAS and LAPACK. The analyses use CPU calculations, not GPU fitting. Four worker processes handle outer samples or genes, with one fitting core per R worker; `OPENBLAS_NUM_THREADS=1` and `OMP_NUM_THREADS=1`. Processes were not pinned to specific physical cores, and the host was not reserved exclusively for benchmarking. Installed memory is machine capacity, not measured peak analysis memory.

The four conditional scenarios each have 500 outer samples and 199 bootstrap refits per sample. Thus 398,000 is the total number of bootstrap margin fits across scenarios, not a number of cells, genes, or independent null experiments. Adding the 2,000 observed fits gives 400,000 fits. The full timed four-scenario execution lasted 3,122.590 seconds (52.043 minutes), including worker orchestration, record writes, and scenario summaries, but excluding initial package loading and protocol file creation. It is one observed elapsed duration, not a median of repeated full executions or a CPU-time measurement.

Conditional task timers include design construction, simulation, the observed fit, all bootstrap fits, and both fixed and refitted statistics; they stop before the individual record is written. Pancreatic task timers include the observed fit, bootstrap loop, RDS record write,

Table 6: Recorded computational workloads and task timers. A task is one outer sample for a conditional scenario, one gene for pancreas, or one gene-model combination for PBMC. Each task has B bootstrap refits. Median task time and the sum of task times are calculated from saved timers. Because tasks ran concurrently, the sum is neither overall wall time nor process CPU time; timing boundaries also differ as detailed below.

Workload	Cells	Tasks	B	Refits	Median task (s)	Sum (min)
Linear / linear	240	500	199	99,500	1.65	13.91
Periodic / linear	240	500	199	99,500	1.59	13.32
Periodic / periodic	240	500	199	99,500	1.81	15.15
Spline / spline	2,087	500	199	99,500	8.12	165.48
Pancreas	2,087	10	1,999	19,990	222.15	36.57
PBMC / shared	24,134	10	9,999	99,990	202.79	33.97
PBMC / patient	24,134	10	9,999	99,990	179.89	29.61

and tail summaries, but exclude input loading and the final CSV write. PBMC timers start after the observed fit and residual calculation; they include the bootstrap loop and result summaries, but exclude data loading, QC, and record serialization. Overall wall times for the pancreatic and PBMC jobs were not separately retained and cannot be recovered by summing these overlapping task durations. These analyses were not rerun to produce this timing summary. Spline outer-task times have median 8.122 seconds and range 7.253–507.031 seconds. The cause of the long elapsed-time outliers was not recorded; the 52-minute duration should not be interpreted as an isolated, repeatable speed benchmark. The recorded Python environment for the PBMC analyses is Python 3.12.4, NumPy 1.26.4, SciPy 1.13.1, and pandas 2.2.2; the pancreatic analysis uses the R configuration documented in H.3. The present hardware inventory does not retrospectively establish the load or affinity of earlier executions, so these rows are not controlled speed comparisons.

Main-text Table 10 was remeasured on the inventoried host on 9 September 2026, using the existing moment-NB implementation with one Python worker and the same single-thread environment settings. Python, NumPy, SciPy, and pandas versions are those above. One $n = 10,000$, 10-gene warm-up is excluded. Five sequential repetitions each traverse $n = 1,000$, 10,000, 100,000, and 1,000,000 with 10 genes, using seeds $20260909 + r$ for repetition $r = 1, \dots, 5$. Counts are generated and genes processed one at a time. Separate timers cover count generation, moment fitting, randomized residuals and KS calculation, and the corresponding operations for one bootstrap data set. Table 10 reports the separate median times; the serial projection is $(\text{median } t_{\text{obs}} + 199 \text{ median } t_{\text{refit}})/60$ minutes. Imports, warm-up, output, and analysis setup are excluded from these per-pass timers. The projection is not a measured 199-refit run, and the $24n$ -byte array baseline is not peak RSS. This lightweight feasibility check does not validate a million-cell scDesign3 bootstrap.

Hardware inventory, individual task times, all 20 lightweight timing records, medians, ranges, and environment metadata are retained in `results/computational_reporting/`. The reporting and timing script is `scripts/report_computational_costs.py`; its default invocation rebuilds the tables from saved timings without rerunning the benchmark. The older lightweight timings in the combined historical results file are preserved, but are superseded by this hardware-documented run for Table 10. Rejection rates and real-data diagnostic results are unchanged.

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